Samarth Pratap Singh

Software Engineer — Backend · AI Applications · Full-Stack · Python
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EDUCATION

Vellore Institute of Technology **2023 – 2027**
B.Tech - Computer Science and Engineering - ***CGPA - 8.61/10*** *Bhopal, Madhya Pradesh*

TECHNICAL SKILLS

Languages: Python, C++, SQL, TypeScript, JavaScript
Backend & Databases: FastAPI, Pydantic, REST APIs, PostgreSQL, MongoDB, Redis, Celery
AI & Retrieval: LangGraph, MCP, RAG, Qdrant, Hybrid Search, Reranking, PyTorch, Hugging Face
Testing & Deployment: pytest, GitHub Actions CI/CD, Docker, Azure Container Apps, LangSmith
Tools & Frontend: Git, Linux, Bash, Next.js

EXPERIENCE

AI/ML Engineering Intern **May 2026 – Present**
AmberFlux EdgeAI Private Limited | Remote *Hyderabad, Telangana*

- Built a PDF vision-extraction pipeline (GPT-5): cut processing from ∼7 min to ∼90 sec on 20-page drawings (∼4.7×) via concurrent batch dispatch (`ThreadPoolExecutor` + `asyncio`), retries and schema-enforced outputs.
- Designed a cost-aware extraction flow (regex/Docling first, GPT-5 vision only for missing fields) for project name, builder, site, contractor, architect and dimensions; validated against internal project records at >**0.85** confidence on real lead sheets.
- Built a LangGraph-routed aggregation layer for documents up to **400 pages**, returning validated structured JSON with page-level guardrails and drawing-page exclusion.

PROJECTS

DoCopilot (Multi-Tenant RAG Platform) | Next.js, FastAPI, PostgreSQL, Qdrant, Azure **May 2026**

- Built a multi-tenant RAG platform with JWT auth, tenant-scoped retrieval via Qdrant payload filters, SHA-256 upload dedup, and an async ingestion job state machine (Celery + Redis via Docker Compose).
- Implemented hybrid retrieval (dense MiniLM + BM25, RRF) with cross-encoder reranking (top-20 → top-5) and SSE-streamed, source-cited answers; added regex-based guardrails and per-IP rate limiting.
- Built an evaluation harness (40 questions, separate LLM judge: Llama-3.1-8B judging Qwen3-32B) to compare chunking and retrieval variants: **89.2%** correctness, **90.5%** relevance, **2.86s** avg latency; shipped via GitHub Actions (pytest CI → Docker → ACR → Azure Container Apps).

Argus - Multi-Agent Research Engine | FastAPI, LangGraph, Docker, LangSmith **Feb 2026**

- Built a LangGraph supervisor pipeline (planner, researcher, critic, writer) with LLM-driven `Command(goto)` routing over Tavily, ArXiv and Wikipedia, producing cited reports at **3** depth tiers (∼20/45/90s).
- Isolated the human-in-the-loop interrupt in a dedicated node so routing LLM calls are not repeated when a paused run resumes; capped at **3** critique loops with a 30-min auto-finalize.
- Exposed a FastAPI submit → SSE-stream → fetch API with SQLite/Postgres auto-switching persistence and checkpointing, per-IP rate limiting, LangSmith tracing and Docker packaging.

ContextCore (Stateful Memory Agent) | LangGraph, PostgreSQL, Qdrant, MCP **Jun 2026**

- Built a LangGraph CLI agent with an intent router (task / memory-recall / chat) and conditional edges, executing task and note workflows through a custom FastMCP tool server (create/list/update/delete tasks, save/search notes) backed by PostgreSQL.
- Designed three memory layers (PostgreSQL checkpointing, Qdrant semantic recall, MongoDB user profile) with token streaming via `astream_events` and an async background memory writer so the prompt returns immediately.
- Wrote an LLM-judge eval harness (50 cases): **92%** pass rate; failures traced to tool-call and token-limit errors,fixed via tool-schema hardening.

CERTIFICATIONS & ACHIEVEMENTS

- Solved **315+** problems on LeetCode across data structures and algorithms.
- Deep Learning Specialization (deeplearning.ai); LangChain for LLM Application Development (deeplearning.ai).